Math 242, S13
Exam 3

Name: Answer
SSN: xxx-xx-

Instructions:

- There are totally 4 problems.
- You must show all of your work to receive a credit for a correct answer.

Problem	Points	Score
1	30	
2	25	
3	25	
4	20	
Total	100	

Good Luck !

Problem 1. (30 points) Find the Laplace or inverse Laplace transforms of the following functions:

(a) $\mathcal{L}\{t^3 - 3\cosh 2t\}$

$$= \frac{3!}{s^4} - 3 \cdot \frac{s}{s^2+4}$$

$$= \frac{6}{s^4} - \frac{3s}{s^2+4}$$

(b) $\mathcal{L}\{t^3 e^{4t}\}$ (Hint: $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$ where $F(s) = \mathcal{L}\{f(t)\}$.)

$$= \frac{6}{(s-4)^4}$$

(c) $\mathcal{L}^{-1}\Big\{\frac{1}{s(s^2+9)}\Big\}$ (Hint: $\mathcal{L}^{-1}\big(\frac{F(s)}{s}\big) = \int_0^t \mathcal{L}^{-1}\{F(s)\}d\tau$.)

$$= \int_0^t \mathcal{L}^{-1}\Big\{\frac{1}{s^2+9}\Big\} d\tau$$

$$= \int_0^t \frac{1}{3}\sin 3\tau \, d\tau$$

$$= \Big[-\frac{1}{9}\cos 3\tau\Big]_{\tau=0}^{\tau=t}$$

$$= -\frac{1}{9}\cos 3t + \frac{1}{9}$$

Problem 2. (25 points) Use Laplace transform to solve the initial value problem:

$$x'' - 9x = 0; \quad x(0) = 5, \ x'(0) = 3.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\mathcal{L}\{x''(t)\} = s^2X(s) - s\,x(0) - x'(0)$$
$$= s^2X(s) - 5s - 3$$

Then

$$\mathcal{L}\{x'' - 9x\} = \mathcal{L}\{0\} \text{ gives us}$$
$$\mathcal{L}\{x''\} - 9\mathcal{L}\{x\} = 0$$
$$s^2X(s) - 5s - 3 - 9X(s) = 0$$
$$X(s) = \frac{5s+3}{s^2-9}$$
$$\Rightarrow x(s) = \mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\left\{\frac{5s+3}{s^2-9}\right\}$$
$$= \mathcal{L}^{-1}\left\{\frac{5s}{s^2-9}\right\} + \mathcal{L}^{-1}\left\{\frac{3}{s^2-9}\right\}$$
$$= 5\cosh 3t + \sinh 3t$$

Problem 3. (25 points) Use Laplace transform to solve the initial value problem

$$x'' + 2x' - 8x = 2; \quad x(0) = x'(0) = 0.$$

Solution: Let $\mathcal{L}\{x(t)\} = X(s)$, then

$$\mathcal{L}\{x'(t)\} = sX(s) - x(0) = sX(s)$$

$$\mathcal{L}\{x''(t)\} = s^2X(s) - sx(0) - x'(0) = s^2X(s)$$

Then

$$\mathcal{L}\{x'' + 2x' - 8x\} = \mathcal{L}\{2\}$$

$$\mathcal{L}\{x''\} + 2\mathcal{L}\{x'\} - 8\mathcal{L}\{x\} = \mathcal{L}\{2\}$$

$$s^2X(s) + 2sX(s) - 8X(s) = \frac{2}{s}$$

$$\Rightarrow X(s) = \frac{2}{s(s^2+2s-8)}$$

$$\frac{2}{s(s^2+2s-8)} = \frac{2}{s(s-2)(s+4)} = \frac{A}{s} + \frac{B}{s-2} + \frac{C}{s+4}$$

$$= \frac{A(s-2)(s+4) + Bs(s+4) + Cs(s-2)}{s(s-2)(s+4)}$$

$$= \frac{A(s^2+2s-8) + B(s^2+4s) + C(s^2-2s)}{s(s-2)(s+4)}$$

$$= \frac{(A+B+C)s^2 + (2A+4B-2C)s - 8A}{s(s-2)(s+4)}$$

$$\Rightarrow \begin{cases} A+B+C = 0 \\ 2A+4B-2C = 0 \\ -8A = 2 \end{cases} \Rightarrow \begin{cases} A = -\frac{1}{4} \\ B = \frac{1}{6} \\ C = \frac{1}{12} \end{cases}$$

$$\Rightarrow x(s) = \mathcal{L}^{-1}\left\{\frac{2}{s(s^2+2s-8)}\right\}$$

$$= \mathcal{L}^{-1}\left\{-\frac{1}{4}\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{6}\frac{1}{s-2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{12}\frac{1}{s+4}\right\}$$

$$= -\frac{1}{4} + \frac{1}{6}e^{2t} + \frac{1}{12}e^{-4t}.$$

Problem 4. (20 points)
(a) Let $f(x) = t$ and $g(t) = e^{3t}$, compute the convolution

$$(f * g)(t) = \int_0^t f(\tau)g(t-\tau)\,d\tau.$$

Solution: $(f*g)(t) = \int_0^t \tau e^{3(t-\tau)}\,d\tau$

$$= e^{3t}\int_0^t \tau e^{-3\tau}\,d\tau$$

$$= e^{3t}\left[\left(\tau\cdot\frac{e^{-3\tau}}{(-3)}\right)_{\tau=0}^{\tau=t} - \int_0^t \frac{e^{-3\tau}}{(-3)}\,d\tau\right]$$

$$= e^{3t}\left[-\frac{te^{-3t}}{3} - \left(\frac{e^{-3\tau}}{9}\right)_{\tau=0}^{\tau=t}\right]$$

$$= e^{3t}\left[-\frac{te^{-3t}}{3} - \frac{e^{-3t}}{9} + \frac{1}{9}\right]$$

$$= -\frac{t}{3} - \frac{1}{9} + \frac{e^{3t}}{9}$$

(b) **Theorem of Differential Tansforms** tells us that $\mathcal{L}\{-tf(t)\} = F'(s)$ where $F(s) = \mathcal{L}\{f(t)\}$. Use this formula to compute $\mathcal{L}\{t\cos 4t\}$.

Solution: $\mathcal{L}\{t\cos 4t\} = -\mathcal{L}\{-t\cos 4t\}$

$$= -\frac{d}{ds}\left[\mathcal{L}\{\cos 4t\}\right]$$

$$= -\frac{d}{ds}\left[\frac{s}{s^2+16}\right]$$

$$= -\frac{1\cdot(s^2+16) - s(2s)}{(s^2+16)^2}$$

$$= -\frac{16-s^2}{(s^2+16)^2}$$

$$= \frac{s^2-16}{(s^2+16)^2}$$